G. The Totient Challenge

time limit per test: 2 seconds
memory limit per test: 256 megabytes

In this challenge, n players are arranged in a line with positions numbered from 1 to n. The goal of the challenge is simple: only a few players will survive, and the rest will be eliminated based on their position and a mathematical property known as Euler's Totient Function.

After the players are arranged, a guard will go through each position and determine whether the player in that position should be eliminated or not. The decision to eliminate a player depends on a mathematical condition involving Euler's Totient Function, $\varphi(k)$, which is defined as the number of integers from 1 to k that are coprime with k. To make the decision, the guard will check each position k $(1 \le k \le n)$ and ask, does $\varphi(k)$ divide k? If yes, the player survives; otherwise, the player in that position will be eliminated.

Your task is to determine how many players will survive.

Input

A single integer n $(1 \le n \le 10^{18})$ — number of players.

Output

Output a single integer representing the number of survivors.

Examples

input
```
6
```
output
```
4
```

input
```
1000000
```
output
```
130
```

Note

Euler's totient function, denoted as $\varphi(n)$, counts the number of positive integers from 1 to n that are coprime to n, meaning they do not share any common divisor with n other than 1.

The function satisfies the following formula for any positive integer n with prime factorization:

$$n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$$

where $p_1, p_2, \dots, p_k$ are distinct prime numbers. The totient function is given by:

$$\varphi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\dots\left(1 - \frac{1}{p_k}\right)$$

For a prime number p, we have:

$$\varphi(p) = p - 1$$

since all numbers less than p are coprime to it.

test case 1

- $\varphi(1) = 1$ because only the number 1 is coprime with itself.
- $\varphi(2) = 1$ because only the number 1 is coprime to 2.
- $\varphi(3) = 2$ because only the numbers 1 and 2 are coprime to 3.
- $\varphi(4) = 2$ because only the numbers 1 and 3 are coprime to 3.
- $\varphi(5) = 4$ because only the numbers 1 and 2 are coprime to 3.
- $\varphi(6) = 2$ because only the numbers 1 and 5 are coprime to 6.

So only players in positions $1, 2, 4$ and 6 are the survivors

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Finished

Practice

→ Virtual participation

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Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
323054901	Jun/06/2025 00:30	Accepted